WET LAB REAGENT LOGBOOK

Reagent Preparation, Mass Balance & Quality Control Tracking

Facility: Apex BioLabs Core Facility **Unit:** Buffer & Media Prep Suite B-304

REAG-LOG-2026-B3

Audit Log Ref: AL-88392-X

TOTAL BATCHES LOGGED	QC APPROVED & ACTIVE	PENDING QC VERIFY	EXPIRED / QUARANTINED
142	128	9	5
Quarter Q3 2026	Ready for Assay Use	Hold for Sterility	Flagged Disposal

SECTION 1: MASTER REAGENT PREPARATION RECORD – BATCH # REAG-2026-0891

Reagent Name: 10X Tris-EDTA (TE) Buffer pH 8.0
Batch / Lot ID: **REAG-2026-0891-TE80**
Target Volume: 1000.0 mL (1.0 L)
Prepared By: Dr. Eleanor Vance (Analyst)
SOP Reference: SOP-BUFF-012 Rev 4.0

Preparation Date: 2026-07-24 **Time:** 09:30
Expiration Date: 2027-01-24 (6 Months)
Storage Condition: Room Temp (20°C to 25°C)
Verified By: Marcus Chen (QA Lead)
Quality Grade: Molecular Biology Grade

Formulation & Chemical Mass Calculation Block

Chemical Component	CAS #	MW	Target Conc.	Calc. Mass	Weighed Mass	Vendor & Lot #
Tris Base [Tris(hydroxymethyl)aminomethane]	77-86-1	121.14	100 mM	12.114 g	12.115 g	Synapse #SY-88291
EDTA Disodium Salt Dihydrate	6381-92-6	372.24	10 mM	3.722 g	3.723 g	Nexa #NX-44012
Hydrochloric Acid (HCl, 6M Stock)	7647-01-0	N/A	pH Adj.	~ 4.2 mL	4.18 mL	Apex #AP-90112
Ultrapure Water (Milli-Q 18.2 MΩ·cm)	7732-18-5	18.02	Diluent	Q.S. 1000 mL	982.5 g	In-House #TANK-02

Step-by-Step Preparation Protocol Execution

Step	Procedure Specification	Check	Observed Parameter / Notes
01	Tare analytical balance (Bal-04). Weigh Tris Base (12.114 g) and EDTA (3.722 g).	[✓]	Weighed 12.115 g Tris, 3.723 g EDTA.
02	Transfer solutes into 1000 mL glass beaker. Add 800 mL ultrapure ddH_2O.	[✓]	Dissolved under magnetic stirring at 400 RPM.
03	Calibrate pH meter (pH-02) using pH 4.01, 7.00, and 10.01 standard buffers.	[✓]	Slope = 99.2%, Temp compensation ON (22.4°C).
04	Slowly titrate with 6M HCl under stirring until target pH 8.00 ± 0.05 is achieved.	[✓]	Final titration volume: 4.18 mL 6M HCl. pH = 8.01.
05	Transfer solution to 1000 mL volumetric flask. Q.S. to mark with ddH_2O.	[✓]	Meniscus aligned precisely at 1000.0 mL mark.
06	Filter sterilize through 0.22 µm PES membrane filter into sterile storage bottle.	[✓]	Autoclave indicator strip turned black. Cycle #402.

Quality Control, Verification & Sign-Off

Final Measured pH: 8.01 at 22.5°C (Spec: 8.00 ± 0.05)
Physical Appearance: Clear, colorless, particle-free
DNase/RNase Test: Negative (Fluorescence Assay Pass)
Analyst Signature: *Dr. Eleanor Vance*

Conductivity: 14.2 mS/cm at 23.0°C
Sterility Check: Passed (0.22 µm Filtered + Autoclaved)
Refractometer Index: 1.3342 ± 0.0005
QA Approval: *Marcus Chen, Lead QA*

REAGENT BATCH IDENTIFIER BARCODE LOG

Scan barcode for automated LIMS inventory entry and lot genealogy tracking.

REAG-2026-0891-TE80